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CBASS phage defense and evolution of antiviral nucleotide signaling

Brianna Duncan-Lowey[1,2] and Philip J Kranzusch[1,2,3]

Cyclic oligonucleotide-based antiphage signaling system (CBASS) immunity is a widespread form of antiphage defense in bacteria and archaea. Each CBASS operon encodes a cGAS/DncV-like Nucleotidyltransferase (CD-NTase) enzyme that synthesizes a nucleotide second messenger in response to viral infection. An associated Cap effector protein then binds the nucleotide signal and executes cell death to destroy the host cell and block phage propagation. Here we build upon recent advances to establish rules controlling each step of CBASS activation and antiphage defense. Comparative analysis of CBASS, CRISPR, Pycsar, and cGAS-STING immunity provides insight into the evolution of phage defense and animal innate immunity and highlights new questions emerging in the role of nucleotide second messenger signaling in host–virus interactions.

Addresses
[1] Department of Microbiology, Harvard Medical School, Boston, MA 02115, USA

[2] Department of Cancer Immunology and Virology, Dana-Farber Cancer Institute, Boston, MA 02115, USA
[3] Parker Institute for Cancer Immunotherapy at Dana-Farber Cancer Institute, Boston, MA 02115, USA

Corresponding author:
Kranzusch, Philip J (philip_kranzusch@dfci.harvard.edu)

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Introduction

Cyclic oligonucleotide-based antiphage signaling system (CBASS) immunity is a widespread form of antiviral defense in bacteria that inhibits phage replication [1••]. Following infection, activation of CBASS immunity triggers rapid cell death and results in an abortive infection response that limits viral spread within the bacterial population [1••,2••,3]. More than 5000 CBASS operons have been identified in diverse bacteria and archaea, with each system responding to phage through synthesis of a specialized nucleotide second messenger signal [1••,4••]. CBASS operons are found in genomes of many prokaryotes, demonstrating a broad importance in enabling bacteria to resist phage replication.

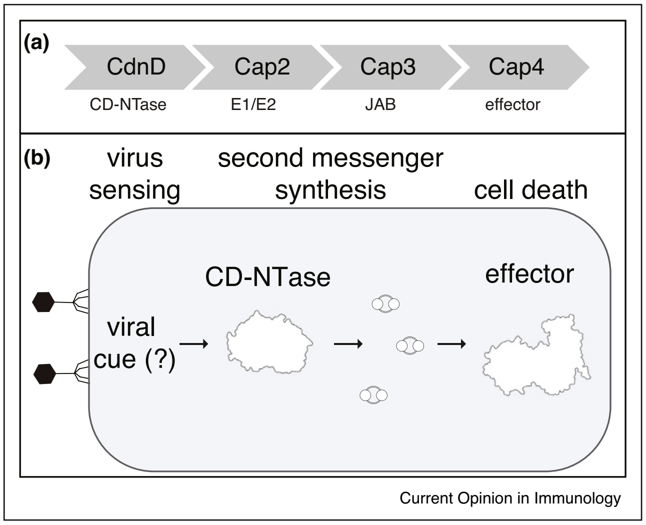
CBASS operons function as self-contained antiviral defense systems that typically encode two to four protein components (Figure 1a) [1••,5•]. All CBASS operons contain a cGAS/DncV-like Nucleotidyltransferase (CD-NTase) enzyme that senses phage replication and catalyzes synthesis of a nucleotide second messenger signal to initiate antiviral defense (Figure 1b) [4••]. CD-NTases are named according to clade designations A–H, for example the CBASS operon in *Enterobacter cloacae* encodes the enzyme CD-NTase in clade D (CdnD). Each CBASS operon also encodes a CD-NTase-associated protein (Cap) effector that functions as a receptor to specifically recognize the CD-NTase nucleotide second messenger signal [2••,5•,6•]. Upon nucleotide-binding, CBASS Cap effectors become activated and induce cell death through degradation of nucleic acid, depletion of essential cellular metabolites, or direct targeting of the host cell membrane [2••,6•,7••,8•,9]. Cap protein names are numbered according to domain organization; for example, the effector family containing a restriction endonuclease-like domain and a SAVED domain is called Cap4. Some CBASS operons contain additional accessory *Cap* genes that are hypothesized to regulate CD-NTase function (Figure 1a) [1••,4••,5•].

Here we summarize recent advances that explain the mechanism of CBASS immune defense. We define general principles that control each step of CBASS signaling including CD-NTase activation, nucleotide second messenger synthesis, and downstream effector function. Surprisingly, structural analysis of CD-NTases and Cap effectors in CBASS immunity has revealed remarkable homology with components of CRISPR antiphage defense and animal cGAS-STING innate immune signaling [2••,4••,7••,8•,10]. We highlight features of CBASS immunity that are shared among other signaling systems and describe the evolutionary roots that connect antiviral immunity in bacteria and animal cells.

CD-NTase enzymes

CD-NTase proteins are bacterial and archaeal signaling enzymes that are present in all CBASS operons [4••,5•]. The CD-NTase family is named after the *Vibrio cholerae* enzyme Dinucleotide cyclase in *Vibrio* (DncV) [11] and the mammalian cytosolic DNA sensor cyclic GMP–AMP synthase (cGAS) [12]. The discovery that CBASS CD-NTases are structural and functional homologs of human

Figure 1



Overview of cyclic oligonucleotide-based antiphage signaling systems (CBASS).
(a) Representative CBASS operon from the bacterium *Acinetobacter baumannii*. Each CBASS operon encodes a CD-NTase (e.g. CdnD) and an effector protein (e.g. Cap4). CBASS operons also frequently encode accessory proteins including Cap2 and Cap3.
(b) Schematic overview of CBASS signaling. Upon phage infection, CD-NTases are activated to produce nucleotide second messengers. The nucleotide second messenger signals by directly binding to the effector protein and inducing a cell death response that prevents phage propagation.

cGAS and animal cGAS-like receptors (cGLRs) demonstrates a direct evolutionary link between human innate immunity and bacterial antiphage defense [1••,4••,10].

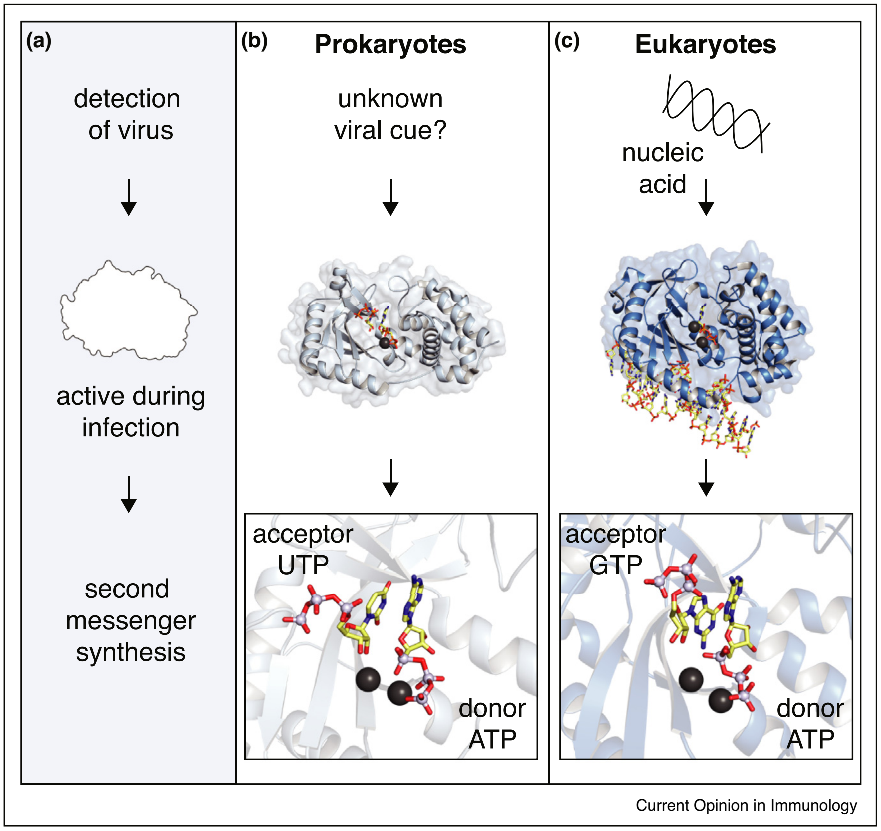
In response to phage infection, bacterial CD-NTases catalyze synthesis of a nucleotide second messenger signal to initiate downstream antiphage defense, a feature common to all CBASS (Figure 2a). CD-NTase nucleotide second messenger synthesis is known to be induced upon phage infection [1••], but the viral cue responsible for CD-NTase activation in CBASS immunity is currently unknown. In human cells, cGAS binds directly to double-stranded DNA mislocalized in the cell cytosol and undergoes a conformational change that results in enzyme activation (Figure 2c) [13–15]. Other cGLRs including *Drosophila* cGLR1 sense double-stranded RNA, demonstrating that animal cGAS-like enzymes are capable of responding to diverse nucleic acid ligands [16,17]. However, bacterial CD-NTase enzymes do not respond to nucleic acid *in vitro* [4••,10], and the lack of spatial compartmentalization in bacteria suggests CBASS immunity does not function through recognition of mislocalized DNA. In contrast to cGAS, many bacterial CD-NTases are robustly active *in vitro* without the addition of an activating ligand [4••]. Additionally, structures of *apo* bacterial CD-NTases closely resemble the active conformation observed in cGAS bound to double-stranded DNA (Figure 2b,c) [4••,10,18,19]. These results are consistent with a model where CBASS immune systems may be held in an inactive state by an inhibitory molecule that represses CD-NTase activation in the absence of phage infection [2••,3,4••,10,18,20]. Defining the activating cue that initiates CD-NTase signaling in bacteria is a major open question in CBASS immunity.

Bacterial CD-NTases and animal cGLRs share a cage-like architecture formed by an N-terminal nucleotidyltransferase core and a C-terminal α-helix bundle (Figure 2b,c bottom) [4••]. The CD-NTase active site functions as a monomeric unit and catalyzes phosphodiester bond formation through a conserved two-metal dependent mechanism [20]. The active site contains distinct nucleotide donor and acceptor pockets with side chains from the enzyme lid that determine nucleotide second messenger product specificity [4••,21]. *Rhodothermus marinus* CdnE, for example, synthesizes the nucleotide second messenger 3′3′-c-UMP–AMP by selecting an ATP molecule in the donor pocket and a UTP molecule in the acceptor pocket, then forming a phosphodiester bond between the 5′ phosphate of the ATP and the 3′ OH of the UTP. The linear intermediate molecule is flipped within the active site and a second phosphodiester bond is added to release the final cyclized product 3′3′-c-UMP–AMP [4••]. cGAS similarly uses a single active site to synthesize the nucleotide second messenger 2′3′-cGAMP (Figure 2c) [13,22–24]. While the synthesis of cyclic dinucleotide products by CD-NTases is well characterized, how this family of enzymes is able to synthesize cyclic trinucleotide nucleotide second messengers remains incompletely understood.

Nucleotide second messenger signals

A key mechanistic aspect common to all CBASS is the use of antiviral nucleotide second messenger signaling to dramatically amplify initial detection into a robust immune response. Activation of a few CD-NTase or cGAS enzymes following viral infection causes multi-turnover production of hundreds of molecules of nucleotide second messenger product (Figure 3a) [4••,13,21]. CD-NTase enzymes in bacteria are extremely diverse at the primary amino-acid level (typically <20% amino acid identity), and many distinct nucleotide second messenger products are used in CBASS immune systems [2••,3,4••,8•,11,21]. The prototypical CD-NTase, *V. cholerae* DncV, synthesizes the cyclic dinucleotide 3′3′-cGAMP composed of two purine bases [10,11]. More recently, bacterial CD-NTases have been characterized that produce pyrimidine-containing signals including 3′3′-c-UMP–AMP and 3′3′-c-UMP–CMP, cyclic trinucleotides including 3′3′3′-c-AMP–AMP–AMP and 3′3′3′-c-AMP–AMP–GMP, and second messengers containing non-canonical 2′–5′ phosphodiester bonds,

Figure 2



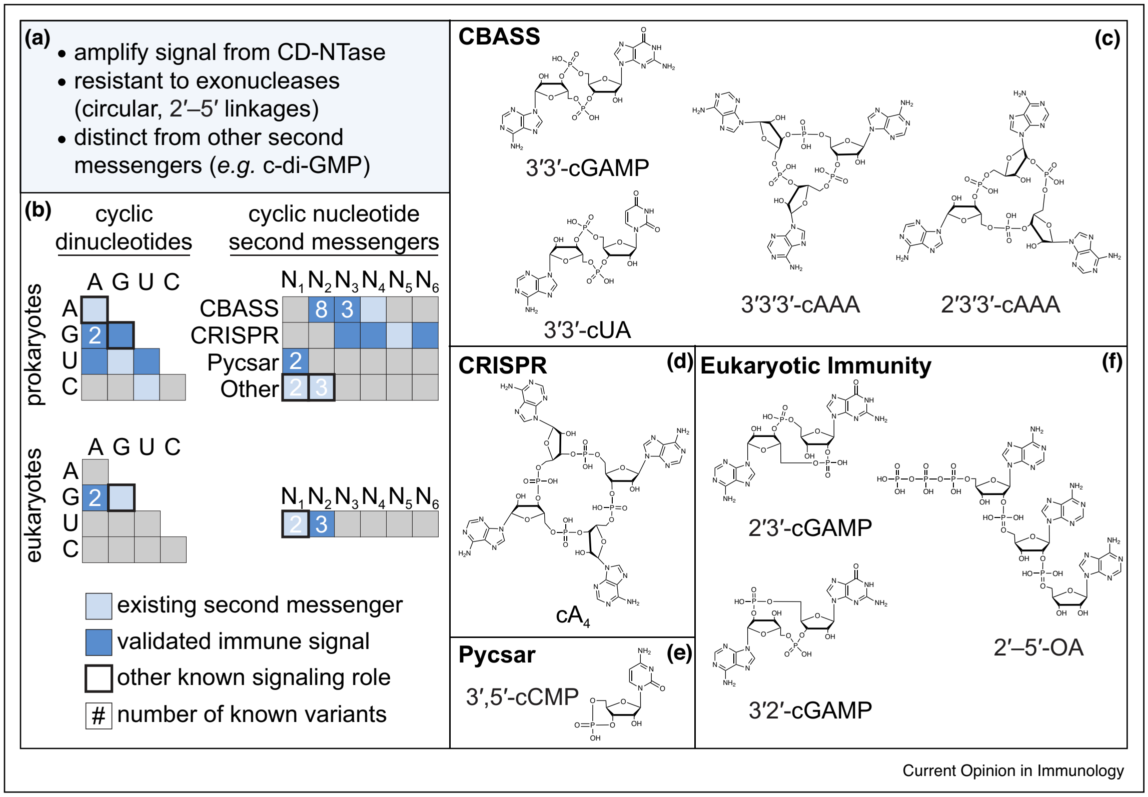
Bacterial CD-NTases.
(a) Overview of CD-NTase function during viral infection. Bacterial CD-NTases sense viral infection and then initiate CBASS defense by producing a nucleotide second messenger. The activating cue sensed during infection is currently unknown.
(b) Cartoon representation of the bacterial CD-NTase *Rm*CdnE (top) that synthesizes 3′3′-c-UMP–AMP using a polymerase-like active site (bottom).
(c) Cartoon representation of human cGAS (top) and the cGAS active site (bottom), highlighting the structural similarity between animal cGAS-like receptors (cGLRs) and bacterial CD-NTase enzymes.

including 2′3′3′-c-AMP–AMP–AMP and 3′2′-cGAMP [2••,3,4••,25] (Figure 3c). Thus far, 7 distinct CD-NTase products have been verified as functional signals in CBASS immunity [2••,3,4••,8•,9,25], and analysis of CD-NTase specificity suggests up to 180 distinct nucleotide and phosphodiester-linkage combinations may exist in biology (Figure 3b,c) [2••].

In addition to CBASS, antiviral nucleotide second messenger signals also function in Pycsar immunity, type III CRISPR systems and in animal cGLR immune signaling. In Pycsar systems, PycC enzymes synthesize cyclic the mononucleotides cUMP and cCMP in response to phage infection (Figure 3e) [26••]. In type III CRISPR, recognition of phage infection results in activation of the protein Cas10 that synthesizes larger cyclic oligoadenylate rings 3–6 nucleotides in size (Figure 3d) [27••,28••]. Similar to activation of CBASS Cap effectors, these CRISPR and Pycsar nucleotide second messengers activate downstream effectors that inhibit phage replication (Figure 3b) [26••,27••,28••,29,30]. Several effectors, including NucC and SAVED-containing effectors, can be found in both CBASS and CRISPR systems [7••,9,31].

In animals, the mammalian protein cGAS synthesizes 2′3′-cGAMP to activate the downstream receptor STING and type I interferon responses [13,22–24]. Recently, 3′2′-cGAMP has been identified both as another metazoan immune nucleotide signal synthesized by the dsRNA sensor cGLR1 in *Drosophila* (Figure 3b,f) [16] and as second messenger in bacterial CBASS [25]. Oligoadenylate synthase 1 (OAS1), is an enzyme structurally related to bacterial CD-NTases and animal cGLRs that senses dsRNA and synthesizes a linear nucleotide second

Figure 3



Nucleotide second messengers.
(a) Overview of shared features of nucleotide second messengers in CBASS.
(b) Summary of known cyclic oligonucleotide second messengers in prokaryotes (top) and eukaryotes (bottom).
(c) Example cyclic oligonucleotides synthesized by bacterial CD-NTases in CBASS operons: 3′3′-cGAMP (*Vc*DncV product), 3′3′-c-UMP–AMP (*Rm*CdnE product), 3′3′3′-c-AMP–AMP–AMP (*Ec*CdnC product), 2′3′3′-c-AMP–AMP–AMP (*Ab*CdnD product).
(d) Example cyclic oligonucleotide synthesized by Cas10 in type III CRISPR systems.
(e) Example cyclic mononucleotide synthesized by PycC in Pycsar immunity.
(f) Nucleotide second messengers known to be involved in metazoan immune signaling: 2′3′-cGAMP (mammalian cGAS and common insect cGLR product), 3′2′-cGAMP (*Drosophila* cGLR1 product), 2′–5′-OA (mammalian OAS1 product).

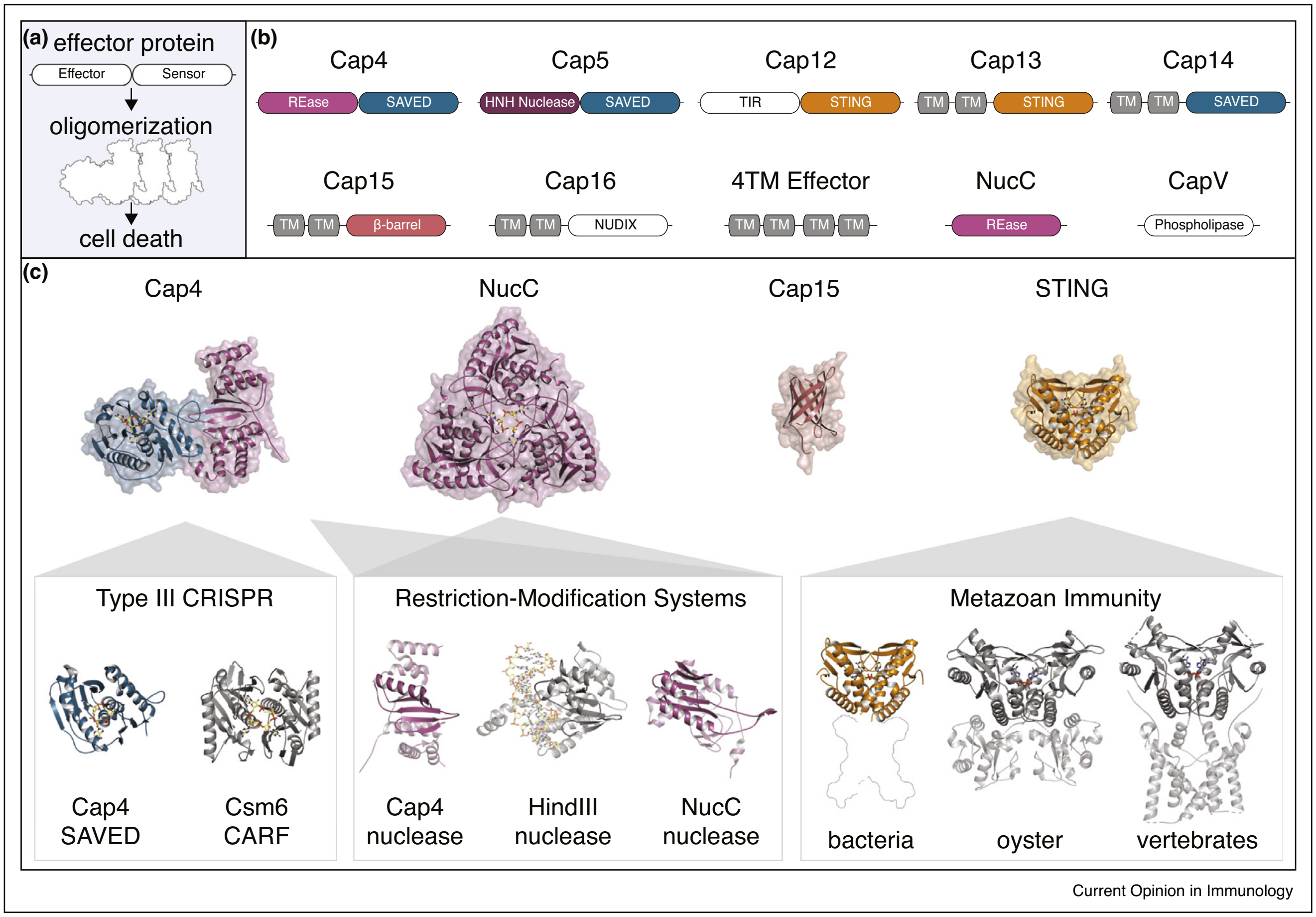
messenger 2′–5′-linked oligoadenylate (2′–5′-OA) (Figure 3b,f) [32]. However, no linear nucleotide second messengers have been identified in bacterial antiphage defense. While nucleotide second messengers, like 2′3′-cGAMP, seem to be restricted to immune signaling in animals, c-di-GMP is used to regulate cell differentiation in the eukaryote *Dictyostelium* [33].

The role of CBASS in controlling cell death and antiphage defense necessitates that CD-NTase nucleotide second messenger products are distinct from signals used during normal bacterial growth. CBASS Cap effectors sensitively respond to 1–10 nM levels of nucleotide second messenger [2••,7••,8•], and cell death responses must therefore be insulated from nucleotide second messenger signals used in normal housekeeping functions including c-di-GMP and c-di-AMP (Figure 3a) [34,35]. For example, c-di-GMP is a rare CD-NTase signaling product that has only been observed in bacteria that lack all other c-di-GMP signaling components [8•]. It is likely that evolution of atypical nucleotide second messenger signals also facilitates horizontal transfer of CBASS operons between distantly related phyla [5•]. Additionally, rapid divergence of nucleotide signals and evolution of noncanonical 2′–5′ phosphodiester linkages may enable CBASS immune systems to overcome viral nuclease enzymes that degrade second messenger signals [36,37•].

CBASS Cap effectors as nucleotide second messenger receptors

CBASS nucleotide second messengers signal by directly binding to and activating a downstream Cap effector. Each Cap effector has two main functions: (1) specific recognition of the nucleotide second messenger produced

Figure 4



CBASS Cap effector proteins.
(a) Overview of CBASS Cap effector function in inducing cell death. Each effector specifically senses a CD-NTase nucleotide second messenger and is then activated typically through oligomerization.
(b) Schematic representation of domain architecture of experimentally characterized CBASS effectors. Common nucleotide sensor domains (e.g. SAVED, STING) can be found fused to various effector domains (e.g. nucleases, TIR, TM).
(c) Cartoon representation of known CBASS effector structures (top) and comparison to evolutionarily related components in other bacterial defense systems and in metazoan innate immunity (bottom).

by the cognate CD-NTase protein in response to phage infection and (2) induction of cell death to prevent phage propagation (Figure 4a). Many Cap effectors are two-domain proteins with a discrete 'sensor' domain dedicated to nucleotide recognition fused to an enzymatic or toxin-like 'effector' domain responsible for executing cell death. The modular architecture of Cap effectors enables diversification of CBASS defense systems including alternative combinations of sensor and effector domain fusions to control cell death (Figure 4b) [2••,7••,8•,9].

Structures of CBASS Cap effectors explain the mechanism of nucleotide second messenger recognition and reveal homology with components of distantly related bacterial and animal antiviral signaling systems. SAVED domains are a common sensor domain in Cap effectors, representing >1600 examples and ~30% of all CBASS operons (Figure 4c) [2••,38•]. The structure of Cap4 (a restriction endonuclease-SAVED fusion) revealed that SAVED domains form a flat surface with individual pockets to coordinate each base of the nucleotide second messenger signal [2••]. The Cap4 structure additionally revealed that SAVED is a fusion of two CARF (CRISPR-associated Rossmann Fold) protein subunits which control recognition of cyclic oligonucleotides in type III CRISPR immunity, demonstrating a direct connection between CBASS and CRISPR immunity [2••,39,40]. Fusion of CARF proteins into a single-chain SAVED receptor enables recognition of diverse CBASS ligands including cyclic trinucleotides and asymmetric CD-NTase products. CBASS and type III CRISPR immune systems share additional effector proteins including an REase-type effector named NucC [7••]. Structures of NucC also revealed that some CBASS effectors do not contain a dedicated nucleotide-binding domain and instead recognize the CD-NTase product through loops emerging from the enzymatic or toxin-like domain [7••]. Although rarer in CBASS immunity (∼100 operons), Cap12 and Cap13 proteins are notable effectors as they contain a nucleotide sensor domain with direct homology to the animal innate immune protein STING [8•]. In both CBASS and animal immunity, STING domains form a homodimeric V-shaped receptor that recognizes cyclic dinucleotide signals in a deep central pocket (Figure 4c) [8•,24].

Several Cap effectors include additional sensor domains that appear to be unique to CBASS immunity. Recently, structures of Cap15 revealed a minimal eight-stranded beta-barrel domain that is responsible for nucleotide second messenger recognition (Figure 4c) [9]. Likewise, bioinformatics analysis has identified an enormous diversity of Cap effectors with many uncharacterized putative nucleotide second messenger sensor domains [38•]. Cap effectors like CapV and Cap16 respond to nucleotide second messengers but do not contain any known nucleotide binding domains [6•,9], suggesting that analysis of CBASS components will continue to reveal new modules capable of selective nucleotide second messenger recognition.

CBASS Cap effectors encode a large variety of toxin-like effector domains to execute cell death. The most common strategy to induce cell death in CBASS is host membrane disruption, either through activation of phospholipases that degrade the membrane [1••,6•] or activation of transmembrane-containing effectors that disrupt the bacterial inner membrane [9]. Many effectors contain a nuclease effector domain that indiscriminately degrades dsDNA upon activation. For example, Cap4 and NucC both contain endonuclease-like domains with homology to bacterial Restriction-Modification system nucleases used to cleave foreign DNA (Figure 4c) [41]. The CBASS nucleases Cap4 and NucC, however, lack the structural features that define sequence specificity and instead degrade dsDNA in a sequence-independent manner [2••,7••]. Additionally, some Cap effectors contain TIR domains that catalyze rapid degradation of the essential metabolite NAD^+ [8•], and more rare effectors have been implicated in CBASS immunity including those with predicted protease and phosphatase effector domain function [2••,5•,38•].

In most cases, Cap effector activation occurs through sensor domain oligomerization where nucleotide second messenger recognition drives higher-order complex assembly. Electron microscopy analysis demonstrates that Cap4 (REase-SAVED) and Cap12 (TIR-STING) form filaments in the presence of the activating nucleotide second messenger signal [2••,8•]. Likewise, nucleotide signal recognition induces oligomerization of the CBASS effector NucC and formation of a stable hexameric state capable of promiscuous DNA degradation [7••]. Ligand-induced oligomerization therefore represents a common activation mechanism among structurally diverse CBASS effectors.

Open questions

While recent advances have uncovered many mechanistic details of CBASS immunity, several fundamental questions remain. First, the molecular mechanisms by which phage infection activates CD-NTases and initiates CBASS defense remain unknown. Individual CBASS operons provide defense to phylogenetically diverse phage, indicating that the activating cue sensed during phage replication is broadly conserved and unlikely to be related to a specific viral protein or nucleic acid sequence [1••,2••,3]. Similarly, CBASS operons frequently encode accessory *Cap* genes, but no mechanistic role is known for the most common accessory proteins including Cap2 (a predicted E1/E2-like ligase) and Cap3 (a predicted deubiquitinase-family peptidase) [5•,38•,42]. Accessory Cap2 and Cap3 proteins are required for efficient defense against some, but not all, phage infections [1••], suggesting a role in modifying the specificity of antiviral defense. Another accessory protein found in some operons, Cap7 (a HORMA-domain containing protein) was demonstrated to directly bind the CBASS operon CD-NTase and contain a potential binding interface for foreign peptide recognition [3], suggesting this family of accessory proteins may play a role in phage sensing. Finally, phage encode evasion proteins to circumvent many bacterial defense systems, including anti-CRISPR proteins and inhibitors of Restriction-Modification systems [43,44]. Discovery of phage anti-CBASS proteins will provide important tools to explain individual steps of pathway activation and further define the host-pathogen interactions that mediate successful CBASS antiphage defense.

Authors' contribution

Review was written by BD-L and PJK, figures were prepared by BD-L.

Conflict of interest statement

Nothing declared.

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- • of special interest
- •• of outstanding interest

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